DHRUV GUPTA

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ABOUT

Applied AI engineer building agents that plan, reason, validate, and execute complex tasks. From autonomous data pipelines to RAG research tools and a language driven robotic arm, turning reasoning into reliable execution. Currently exploring physical AI and robotics.

WORK EXPERIENCE

KYC HUB **Remote**

Software Engineer **Jan 2026 - June 2026**

- Built agentic AI workflows to automate onboarding of global AML data sources (sanctions, criminal records, PEPs), replacing manual scraper development.
- Implemented a two-agent approach consisting of an investigative agent to automatically analyze website DOM structures, craft targeted extraction action plans, define precise attribute schemas, and establish custom validation logic.
- Engineered a datasource coder agent that generates Python scrapers, executes web scraping tasks, validates retrieved data, and seamlessly feeds downstream processing pipelines.
- Orchestrated long running extraction and ingestion pipelines with Temporal for durable execution, converting heterogeneous web data into structured records for compliance systems.

SHORTHILLS AI Gurugram, Haryana

Software Developer Intern **Feb 2025 - Aug 2025**

- Built an AI candidate job matching system using OpenAI embeddings and Elasticsearch semantic retrieval to rank candidates against job requirements, with a recruiter dashboard to review ranked profiles.
- Developed a multi format resume parsing pipeline (LlamaParse, Tesseract OCR) to extract structured data from PDF, DOCX, and scanned documents.
- Automated resume generation in standardized corporate formatting, preserving key layout elements, enhancing content structure, and rewriting project summaries for improved clarity.
- Engineered scalable and modular ingestion workflows using Apache NiFi, integrating end-to-end functionality into Microsoft Teams chat for streamlined user access.

HUGHES COMMUNICATION INDIA Gurugram, Haryana

Software Development Intern June 2024 - Aug 2024

- Built a real-time Python terminal tracking system to monitor active network endpoints across distributed locations.
- Integrated live geospatial REST APIs with SQL spatial queries and an OpenStreetMap interface to automate geofenced access restrictions.
- Developed a CNN/YOLOv8 model to verify dish antenna placement from field imagery, automating quality sign off and cutting manual inspection time.

PROJECTS

AERO | Agentic Execution for Robotic Operations | GitHub | Live

Tech Stack: Python, JavaScript, Three.js, 3D Simulation, Spatial Mapping

- Built a simulation first agentic robotics platform that translates natural language instructions into executable manipulation tasks through language understanding, action planning, and spatial reasoning.

- Implemented a simulated 5 DOF robotic arm with geometric inverse kinematics, supporting block stacking and pyramid assembly.
- Designed a deterministic, local execution pipeline that runs without a cloud AI API at runtime, enabling repeatable experiments.
- Developed spatial mapping and tabletop exploration for object localization, with a real time Three.js studio for visualizing arm motion and execution telemetry.

PaperScope AI | Full-Stack AI Research Assistant | GitHub | Live

Tech Stack: Python, FastAPI, Next.js, React, Google Gemini, Docker

- Built a full stack RAG platform for academic research with PDF analysis, source grounded Q&A, and cross paper synthesis, with page level citations.
- Engineered a configurable ingestion and retrieval pipeline with page aware extraction, user selectable chunking strategies, adjustable top k parameters, and Gemini for embeddings with a TF-IDF fallback for resilience.
- Created a Next.js interface for summaries, methodology and benchmark analysis, BibTeX generation, and Markdown report export.

EDUCATION

DELHI TECHNOLOGICAL UNIVERSITY — Dec 2021 - June 2025

Bachelor of Technology (BTech)

Cumulative GPA: 8.13

DOON PUBLIC SCHOOL — June 2021

CBSE 12th-89% 10th-88%

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, SQL

Artificial Intelligence: Agentic AI, Retrieval Augmented Generation (RAG), Natural Language Processing (NLP), Large Language Models (LLMs), Embeddings, Prompt Engineering, Tool Calling, Google Gemini, Azure OpenAI

Vision & Robotics: Machine Learning, Deep Learning, OpenCV, YOLOv8, Vision Language Action (VLA) Models, Object Detection, Image Processing, Inverse Kinematics, Trajectory Planning, Spatial Mapping, ROS2, Gazebo, Robot Simulation & Control

Backends, Distributed Systems & Frameworks: FastAPI, Flask, Django, REST APIs, Temporal, Next.js, React, Streamlit, Three.js, Selenium, Playwright, Pandas, NumPy

Databases, Vector Stores & Search: MongoDB, Redis, FAISS, Elasticsearch, Semantic Search

Cloud, DevOps & Environments: Microsoft Azure, Amazon AWS, Docker, Git, Linux

OTHER INFORMATION

- **Codeforces:** Max rating **1508 (Specialist)**, handle: incheif
- **1st Place, Solana Track, LNMHacks 5.0:** Part of a team we built **P2P SOL Marketplace**, a Solana Pay based marketplace for businesses; developed the web interface. Devfolio
- **Co Head, Indian Game Theory Society (DTU):** Led the society and organized events at other colleges, including **Hindu College** and **IIT BHU**.